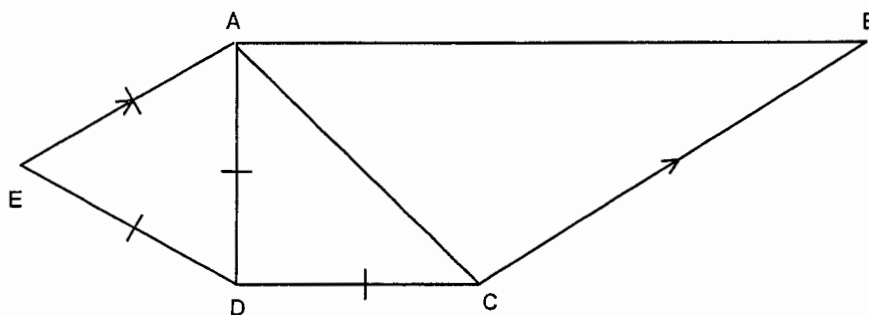


Question 1**8 marks**

- a) Write the equation of the circle with centre $(4, -1)$ and radius 7 units 2
- b) Solve the quadratic equation $x(2x - 3) = 5$ 2
- c) In the diagram below $AE = ED = AD = DC$, $\angle ADC = 90^\circ$ and $AE \parallel BC$.
 $\angle BAC = 51^\circ$



- i) Find the size of $\angle EAB$. Give reasons for your answer. 2
- ii) Find the size of $\angle ABC$. Give reasons for your answer. 2

Question 2 (start a new page)**8 marks**

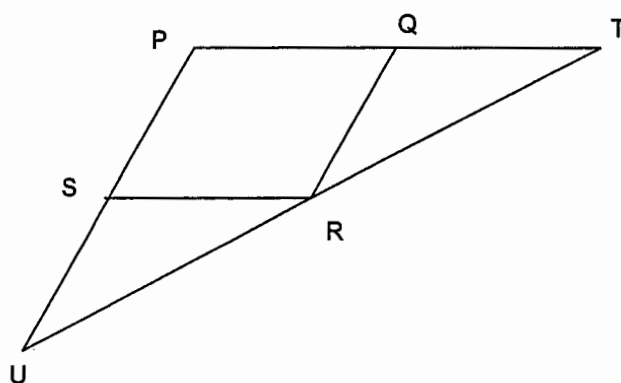
- a) Derive the equation of the locus of a point $P(x, y)$ which moves so as to be equidistant from the two points $A(-2, 4)$ and $B(8, -3)$ 3
- b) For the parabola $(x - 3)^2 = 20(y + 5)$, find:
- i) The coordinates of the vertex 1
 - ii) The focal length 1
 - iii) The coordinates of the focus 1
 - iv) The equation of the directrix 1
 - v) The equation of the axis of symmetry 1

Question 3 (start a new page)**8 marks**

- a) If α and β are the roots of $3x^2 + 4x - 12 = 0$, find without solving the equation:
- | | |
|---|---|
| i) $\alpha + \beta$ | 1 |
| ii) $\alpha\beta$ | 1 |
| iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ | 1 |
| iv) $\alpha^2 + \beta^2$ | 2 |
- b) Find the values of the constants A, B and C such that
- 3
- $$3x^2 - 8x + 6 \equiv A(x-1)^2 + B(x-1) + C$$

Question 4 (start a new page)**8 marks**

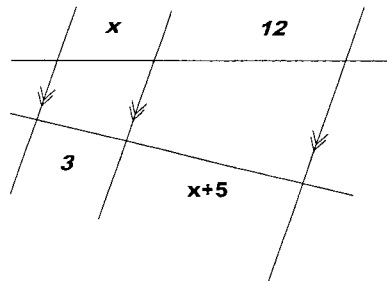
- a) $PQRS$ is a parallelogram. PQ is produced to T so that $QT = QR$ and PS is produced to U so that $SU = PS$. It is now discovered that T , R and U are collinear.
- i) Show that $SU = RQ$ 2
- ii) Prove $PQRS$ is a rhombus 3



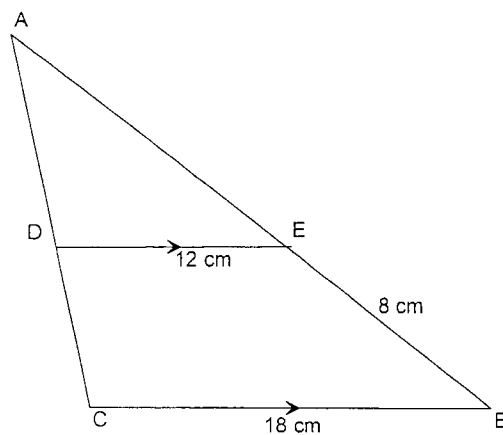
- b) For the function $y = kx^2 - 4\sqrt{3}x + k - 1$,
- i) find an expression for the discriminant. 2
- ii) for what values of k is the function positive definite. 2
- Page 3

Question 5 (start a new page)**8 marks**

- a) Find the axis of symmetry and the vertex of the parabola $y = 5x^2 + 10x + 2$. 2
- b) The roots of the quadratic equation $px^2 - x + q = 0$ are -1, 3. Find p and q. 3
- c) Solve for x 3

**Question 6 (start a new page)****8 marks**

- a) Find the discriminant of the following equation and state the nature of the roots $2x^2 + 3x + 5 = 0$ 2
- b) Solve $2(x^2 + 1)^2 - 19(x^2 + 1) - 10 = 0$ 3
- c) In the diagram below $DE \parallel CB$, $DE = 12$ cm, $CB = 18$ cm and $EB = 8$ cm.



- (i) Prove that $\triangle ADE \parallel \triangle ACB$ 2

- (ii) Find the length of AE . 1

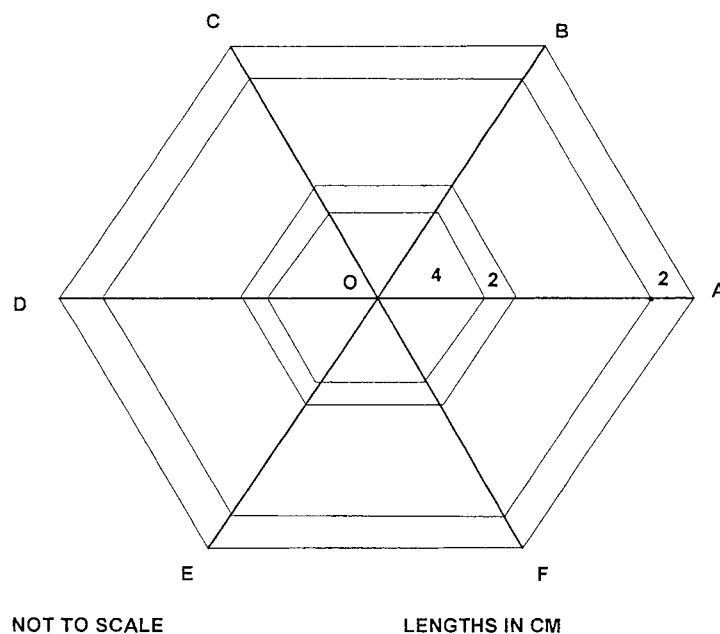
Question 7 (start a new page)**8 marks**

- a) A parabola whose equation is $y = ax^2$, where a is a constant, has the line $y = 12x + 3$ as a tangent.
- i) By equating the two given equations, find a quadratic equation in terms of x and a . **1**
 - ii) By using the discriminant of the quadratic equation found, find the value of a . **2**
 - iii) Find the coordinates of the point of contact between the tangent and the parabola. **2**
 - iv) Sketch the parabola and the tangent line, showing the co-ordinates of intercepts and the point of contact **2**

Question 8 (start a new page)**8 marks**

- a) Write down the formula for:
- i) the n th term of an arithmetic series with first term a and the common difference d 1
 - ii) the sum of the first n terms of this series 1

A particular spider's web consists of a series of regular hexagons with a common centre O , held together by rays through O , as in the figure, where only some of the hexagons are shown.



The vertices of the smallest hexagons are 4cm from O . The vertices of the next hexagons are 2cm further away and they continue at 2cm intervals along the rays until the vertices of the last hexagon ABCDEF are 60cm from O .

- iii) How many hexagons are there? 1
 - iv) What is the length, in cm, of the perimeter of the smallest hexagon? 1
 - v) What is the total length of thread used by the spider in making this web (including the six rays from O)? 1
- b)
- i) If $4^{x+1} = 2^a$ find a 1
 - ii) Hence solve $4^{x+1} - 12(2^x) + 8 = 0$ 2

END OF EXAM



Sydney Technical High School
Assessment Task One
Term 4 2010

Question 1

a) $(x-4)^2 + (y+1)^2 = 49$
 $(x-4)^2 + (y+1) = 7$

b) $x(2x-3) = 5$
 $2x^2 - 3x - 5 = 0$
 $(x+1)(2x-5) = 0$
 $x = -1 \quad x = \frac{5}{2}$

c) i) $\angle EAD = 60^\circ$ (equilateral triangle)
 $\angle DAC = 45^\circ$ (right-angled isosceles)
 $\angle CAB = 51^\circ$ (given)
 $\therefore \angle EAB = 156^\circ$

ii) $\angle EAB + \angle ABC = 180^\circ$ (interior angles)
 $156^\circ + \angle ABC = 180^\circ$ (in // lines)
 $\angle ABC = 24^\circ$ ($AE \parallel BC$)

Question 2

a) $P(x, y)$
 $A(-2, 4)$
 $B(8, -3)$ $PA^2 = PB^2$
 $(x+2)^2 + (y-4)^2 = (x-8)^2 + (y+3)^2$
 $x^2 + 4x + 4 + y^2 - 8y + 16 = x^2 - 16x + 64 + y^2 + 6y + 9$

b) $(x-3)^2 = 20(y+5)$
i) vertex $(3, -5)$
ii) focal length = 5
iii) focus $(3, 0)$
iv) directrix $y = -10$
v) $x = 3$

Question 3

a) $3x^2 + 4x - 12 = 0$

i) $\alpha + \beta = -\frac{4}{3}$

ii) $\alpha\beta = \frac{-12}{3} = -4$

iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-\frac{4}{3}}{-4} = \frac{1}{3}$

iv) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= (-\frac{4}{3})^2 - 2(-4) = 9\frac{1}{9}$

b) $3x^2 - 8x + 6 \equiv A(x-1)^2 + B(x-1) + C$

R.H.S. $= A(x-1)^2 + B(x-1) + C$
 $= A(x^2 - 2x + 1) + Bx - B + C$
 $= Ax^2 - 2Ax + Bx + A - B + C$
 $A = 3$

$-8 = -2A + B$

$-8 = -2(3) + B$

$B = -2$

$A - B + C = 6$

$-3 - (-2) + C = 6$

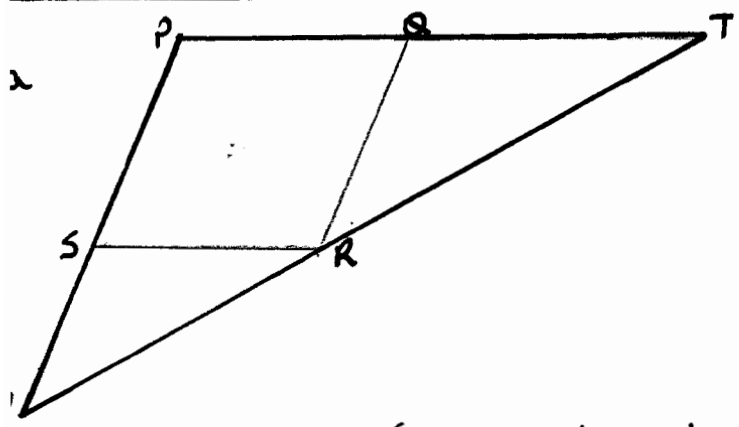
$C = 1$

$A = 3$

$B = -2$

$C = 1$

Question 4



i) $\angle RSU = \angle QPS$ (corresponding $\angle$'s
 $SR \parallel PT$)
 $\angle TQR = \angle QPS$ (corresponding $\angle$'s
 $QR \parallel PU$)

$\therefore \angle TQR = \angle RSU$
 $QR = PS$ (opp sides of parallelogram)
 $SU = PS$ (given).
 $QR = SU$

ii) In $\triangle TQR$ & RSU
 $\angle TQR = \angle RSU$ (proven above)
 $QR = SU$ (proven above)
 $\angle RTQ = \angle URS$ (corresponding $\angle$'s
 $PT \parallel SR$)

$\therefore \triangle TQR \cong \triangle RSU$ (AAS)
 ie $QT = SR$
 $QT = QR$
 $QR = SR$

(adjacent sides of the parallelogram PQRS)

$\therefore PQRS$ is a rhombus

b i) $y = kx^2 - 4\sqrt{3}x + k - 1$
 $\Delta = b^2 - 4ac$
 $\Delta = (-4\sqrt{3})^2 - 4 \times k(k-1)$
 $\Delta = 48 - 4k^2 + 4k$

ii) positive definite
 $a > 0 \quad 48 + 4k - 4k^2 < 0$
 $\Delta < 0 \quad 4k^2 - 4k - 48 > 0$
 $k^2 - k - 12 > 0$
 $(k+3)(k-4) > 0$
 $k < -3 \quad k > 4$
 only solution $k > 4$

Question 5

a) $y = 5x^2 + 10x + 2$
 $x = -10/2 \times 5 = -1$
 vertex $(-1, -3)$

b) $px^2 - x + q = 0$
 $x^2 - \frac{1}{p}x + \frac{q}{p} = 0$

$\alpha + \beta = \frac{1}{p}$

$\alpha + \beta = -1 + 3 = \frac{1}{p}$

$p = \frac{1}{2}$

$\alpha\beta = \frac{q}{p} = -1 \times 3 = \frac{q}{\frac{1}{2}}$
 $-3 = q \times \frac{1}{2}$
 $q = -\frac{3}{2}$

$p = \frac{1}{2}$
 $q = -\frac{3}{2}$

$$c) \quad \frac{x}{12} = \frac{3}{x+5}$$

$$x^2 + 5x - 36 = 0$$

$$(x+9)(x-4) = 0$$

$$x = -9, 4 \quad \text{but } x > 0$$

$$\therefore x = 4$$

Question 6

$$a) \quad 2x^2 + 3x + 5 = 0$$

$$\Delta = b^2 - 4ac$$

$$\Delta = 3^2 - 4 \times 2 \times 5$$

$$\Delta = -31 < 0$$

$$\therefore \Delta < 0 \quad a > 0$$

No real roots

Positive definite.

$$b) \quad 2(x^2+1)^2 - 19(x^2+1) - 10 = 0$$

$$\text{let } m = x^2 + 1$$

$$2m^2 - 19m - 10 = 0$$

$$(2m+1)(m-10) = 0$$

$$m = -\frac{1}{2} \quad m = 10$$

$$x^2 + 1 = -\frac{1}{2}$$

$$x^2 + 1 = 10$$

$$x^2 = -\frac{3}{2}$$

$$x^2 = 9$$

No Solution

$$x = \pm 3$$

$$c) \quad \text{In } \triangle ADE \text{ \& } \triangle ABC$$

$\angle A$ is common

$\angle AED = \angle ABC$ (corresponding $\angle$'s)

$\angle EDA = \angle ACB$ (in parallel lines)

$\therefore \triangle ADE \parallel \triangle ABC$ (equiangular)

$$c) \quad \text{let } AE = x$$

$$\frac{AB}{AE} = \frac{CB}{DE}$$

$$\frac{x+8}{x} = \frac{18}{12}$$

$$12x + 96 = 18x$$

$$6x = 96$$

$$x = 16$$

$$AE = 16 \text{ cm}$$

Question 7

$$a) \quad \begin{aligned} y &= ax^2 \\ y &= 12x + 3 \end{aligned}$$

$$\begin{aligned} ax^2 &= 12x + 3 \quad \text{or} \\ ax^2 - 12x - 3 &= 0 \end{aligned}$$

ii) Since the line is a tangent to the parabola (one point of contact) the roots are equal

$$\Delta = 0 \quad (-12)^2 - 4 \times a \times -3 = 0$$

$$144 + 12a = 0$$

$$144 = -12a$$

$$a = -12$$

iii) Point of Contact $a = -12$

$$ax^2 - 12x - 3 = 0$$

$$-12x^2 - 12x - 3 = 0$$

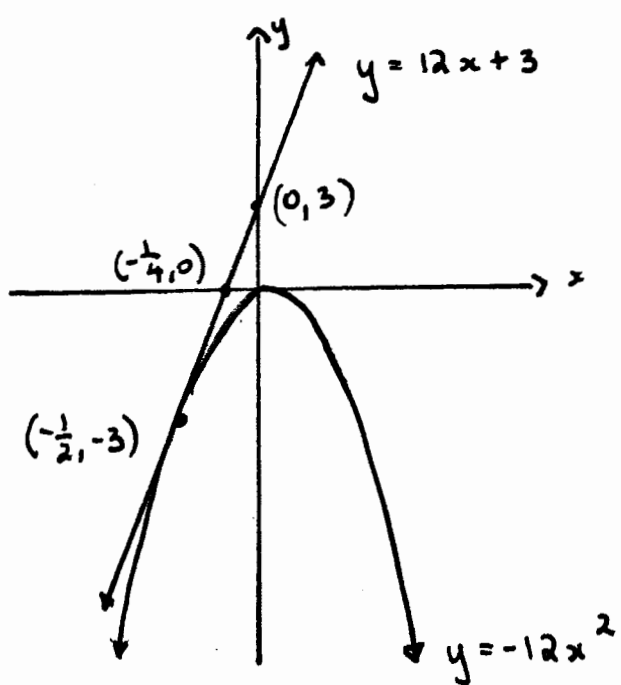
$$4x^2 + 4x + 1 = 0$$

$$(2x+1)^2 = 0$$

$$x = -\frac{1}{2}$$

$$y = -3$$

iv)



v) Sum of hexagon perimeters
= (6x4) + (6x6) + (6x8) + ... + (6x60)
= 24 + 36 + 48 + ... + 360

$$\begin{aligned} a &= 24 & S_n &= \frac{29}{2} [2 \times 24 + (29-1) \times 12] \\ d &= 12 & &= 14.5 \times 336 \\ n &= 29 & &= 5568 \text{ cm} \end{aligned}$$

length of 6 rays $6 \times 60 = 360$

$\therefore$ Total length $5568 + 360 = 5928 \text{ cm}$

Question 8

ai) $T_n = a + (n-1)d$

ii) $S_n = \frac{n}{2} [2a + (n-1)d]$

iii) $a=4$ $4, (4+2), (4+2+2), \dots, 60$
 $d=2$ $4, 6, 8, \dots, 60$
 $T_n=60$ $60 = 4 + (n-1)2$
 $60 = 4 + 2n - 2$
 $58 = 2n$
 $n = 29$

$\therefore$ There are 29 hexagons

iv) Each regular hexagon can be divided into 6 equilateral triangles. The sides of the smallest regular hexagon are 4cm. Perimeter of smallest regular hexagon is $6 \times 4 = 24 \text{ cm}$

b) i) $4^{x+1} = 2^a$
 $2^{2(x+1)} = 2^a$
 $a = 2x + 2$

$$\begin{aligned} 4^{x+1} - 12(2^x) + 8 &= 0 \\ (2^2)^{x+1} - 12(2^x) + 8 &= 0 \\ 2^{2x+2} - 12 \cdot 2^x + 8 &= 0 \\ (2^x)^2 \cdot 2^2 - 12(2^x) + 8 &= 0 \\ \text{let } m &= 2^x \\ 4m^2 - 12m + 8 &= 0 \\ (2m-4)(2m-2) &= 0 \\ m=2 & \quad m=1 \\ 2^x=2 & \quad 2^x=1 \\ x=1 & \quad x=0 \end{aligned}$$